Chapter 5

Markov Chains

Contents

5.1 Definitions and basic properties 59
5.2 Transition matrices and the Chapman–Kolmogorov equations . . 60
5.3 Classification of states 61
5.4 Stationary and limiting distributions 62
5.5 Ergodic theorems 63
5.6 Reversibility and detailed balance 64

Chapters 2 and 3 drew **independent** values: from a known distribution, through an envelope (accept–reject), or from a proposal followed by reweighting (e.g., importance sampling).
All three methods struggle in high dimensions, because a single fixed proposal rarely matches the shape of the posterior. In this chapter, we introduce a different kind of randomness: a sequence in which each value is a random move from the previous one.
Such a sequence is called a **Markov chain**. We are thus interested in knowing when a chain settles into a fixed distribution, how fast it does so, whether a single long trajectory can replace independent draws, and how to build a chain that settles into a distribution of our choice. This knowledge comprises the foundation of the Markov chain Monte Carlo (MCMC) methods, which we will study in Chapter 6.

5.1 Definitions and basic properties

Example 5.1 (Monthly brand choice)**.** Each month, a retailer's customer buys our brand (O), the competitor's brand (K), or nothing (N). Purchase data suggest that next month's choice depends on this month's choice, but not on earlier months. A loyal customer of O stays with O with probability 0.80, switches to K with probability 0.15, and buys nothing with probability 0.05. The three rows of probabilities form the **transition matrix**

$$P = \begin{pmatrix} 0.80 & 0.15 & 0.05 \\ 0.20 & 0.70 & 0.10 \\ 0.30 & 0.30 & 0.40 \end{pmatrix} \begin{matrix} \text{from O} \\ \text{from K,} \\ \text{from N} \end{matrix}$$

with columns in the same order (to O, to K, to N). Each row sums to one (Figure 5.1). The probability of a sequence of choices is a product of transition probabilities: starting from O, the probability of the path O → O → K → O is $P_{\mathrm{OO}}P_{\mathrm{OK}}P_{\mathrm{KO}} = 0.80 \times 0.15 \times 0.20 = 0.024$. A simulated customer over two years reads OOOOOOOKOOONNNNKKKKKKKKKO: long spells in one brand (i.e., state) and occasional switches.

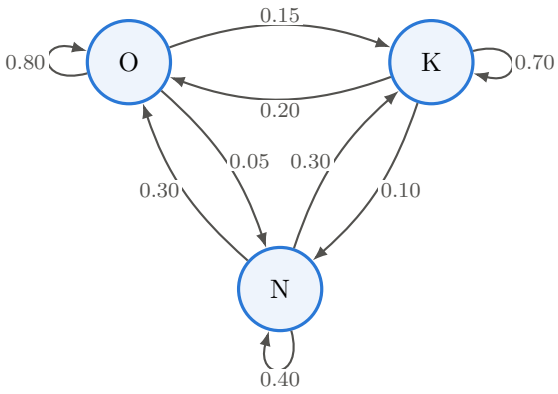


Figure 5.1: The brand-choice chain of Example 5.1: our brand (O), the competitor's (K), no purchase (N). Arrows carry the monthly transition probabilities.

Definition 5.1 (Markov chain)**.** A sequence of random variables $X_0, X_1, X_2, \dots$ taking values in a set $\mathcal{S}$ (i.e.,the **state space**) is a **Markov chain** if, for every t and every states $i_0, \dots, i_{t-1}, i, j$,

$$\Pr(X_{t+1} = j \mid X_t = i, X_{t-1} = i_{t-1}, \dots, X_0 = i_0) = \Pr(X_{t+1} = j \mid X_t = i) = P_{ij}. \quad (5.1)$$

The **Markov property** (5.1) says that the future depends on the past only through the present. The chain is **time-homogeneous** when P_{ij} does not depend on t. On a finite state space, $P = (P_{ij})$ is the **transition matrix**: its entries are non-negative, and each row sums to one. The distribution of X_0 is the **initial distribution** μ_0.

The initial distribution and the transition matrix determine the whole process: $\Pr(X_0 = i_0, X_1 = i_1, \dots, X_t = i_t) = \mu_0(i_0) P_{i_0 i_1} \cdots P_{i_{t-1} i_t}$. Simulating a chain only requires drawing X_{t+1} from row X_t of P, by the discrete inverse transform of Example 2.2. On a continuous state space, the rows of P are replaced by a **transition kernel** $K(x, \cdot)$, the conditional distribution of X_{t+1} given $X_t = x$ (Example 5.6).

5.2 Transition matrices and the Chapman–Kolmogorov equations

Example 5.2 (Where are today's loyal customers in n months?)**.** Suppose every customer buys our brand this month, $\mu_0 = (1, 0, 0)$. Next month, the distribution of choices is the first row of P, $\mu_1 = (0.80, 0.15, 0.05)$. The month after, a customer is in O after a path O→O→O, O→K→O, or O→N→O: $\mu_2(\mathrm{O}) = 0.80 \times 0.80 + 0.15 \times 0.20 + 0.05 \times 0.30 = 0.685$, the first entry of $\mu_1 P$. Repeating,

Month n	O	K	N
0	1.0000	0.0000	0.0000
1	0.8000	0.1500	0.0500
2	0.6850	0.2400	0.0750
5	0.5575	0.3428	0.0997
10	0.5284	0.3667	0.1049
20	0.5263	0.3684	0.1053

Loyalty erodes quickly, then the shares stop changing.

Proposition 5.1 (Chapman–Kolmogorov)**.** *The n-step transition probabilities $P_{ij}^{(n)} = \Pr(X_{t+n} = j \mid X_t = i)$ are the entries of the matrix power P^n, and for all $m, n \geq 0$,*

$$P_{ij}^{(m+n)} = \sum_{k \in \mathcal{S}} P_{ik}^{(m)} P_{kj}^{(n)}, \qquad \text{that is,} \qquad P^{m+n} = P^m P^n. \tag{5.2}$$

Consequently, the distribution of X_n is $\mu_n = \mu_0 P^n$ (i.e., a row vector times a matrix).

Proof. Partition on the state $X_{t+m} = k$ and use the Markov property: $\Pr(X_{t+m+n} = j \mid X_t = i) = \sum_k \Pr(X_{t+m} = k \mid X_t = i) \Pr(X_{t+m+n} = j \mid X_{t+m} = k)$. The matrix form follows by induction from $P^{(1)} = P$. □

Example 5.3 (The customer journey on the website)**.** Within a session, a visitor moves between Browse pages (B), Product pages (V), and the Cart (C), until the session ends with a Purchase (P) or by Leaving (L), as depicted in Figure 5.2. P and L are **absorbing**: once there, the chain stays. Ordering the states as B, V, C, P, L, the transition matrix has the block form

$$P = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix}, \qquad Q = \begin{pmatrix} 0.30 & 0.35 & 0 \\ 0.25 & 0.15 & 0.20 \\ 0.05 & 0.25 & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} 0 & 0.35 \\ 0 & 0.40 \\ 0.40 & 0.30 \end{pmatrix},$$

where Q contains the moves between pages and R the moves that end the session. By Proposition 5.1, the probability of having purchased within n clicks is an entry of P^n: starting from B, it is 0 for one or two clicks (the cart must be reached first), 0.028 for three, 0.049 for five, and 0.059 for ten. The expected number of visits to each page before absorption is given by the **fundamental matrix**

$$N = (I - Q)^{-1} = I + Q + Q^2 + \cdots = \begin{pmatrix} 1.706 & 0.746 & 0.149 \\ 0.554 & 1.493 & 0.299 \\ 0.224 & 0.410 & 1.082 \end{pmatrix},$$

since $(Q^n)_{ij}$ is the probability of being on page j after n clicks without having left. A visitor arriving on B views $1.706 + 0.746 + 0.149 = 2.60$ pages on average, and ends with a purchase with probability $(NR)_{\mathrm{B,P}} = 0.0597$: the conversion rate of about 6% of Chapter 1 emerges from the structure of the website. A simulation of 20 000 sessions gives us 0.057.

5.3 Classification of states

The brand chain and the journey chain behave very differently in the long run. In the first, every customer keeps moving among all three states; in the second, every visitor eventually leaves the pages for good. The following definitions name the difference.

Definition 5.2 (Communication, irreducibility, recurrence)**.** State j is **accessible** from i if $P_{ij}^{(n)} > 0$ for some $n \geq 0$; states i and j **communicate** if each is accessible from the other. Communication splits $\mathcal{S}$ into **classes**. A chain with a single class is **irreducible**. A state i is **recurrent** if, starting from i, the chain returns to i with probability one, and **transient** otherwise.

In the brand chain, every state can reach every other: the chain is irreducible, and being finite, all its states are recurrent. In the journey chain, the classes are $\{\mathrm{B}, \mathrm{V}, \mathrm{C}\}$, $\{\mathrm{P}\}$ and $\{\mathrm{L}\}$. The pages are transient: the probability of ever returning to a page i after leaving it is $1 - 1/N_{ii}$,

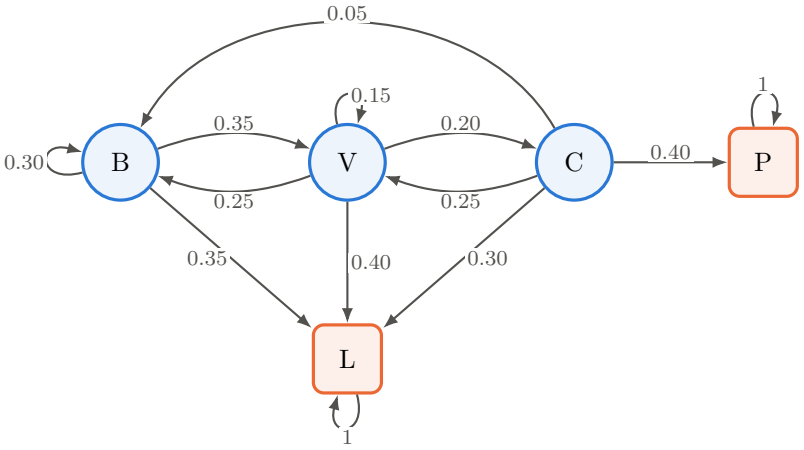


Figure 5.2: Customer journey within a session (Example 5.3): Browse (B), Product page (V), Cart (C), and the absorbing states Purchase (P) and Leave (L).

which equals 0.41 for B, 0.33 for V and 0.08 for C. The absorbing states are recurrent. In a finite chain, a transient state is visited only finitely many times; in an irreducible finite chain, every state is visited infinitely often.

Definition 5.3 (Period)**.** The **period** of a state i is the greatest common divisor of the times $n \geq 1$ with $P_{ii}^{(n)} > 0$. All states of a class have the same period. A chain whose states have period one is **aperiodic**.

Example 5.4 (Alternating promotions)**.** A store alternates a promotion week and a normal week. As a two-state chain, $P = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$: every state returns only at even times, so the period is 2. Starting from a promotion week, $\Pr(\text{promotion at week } n)$ is $1, 0, 1, 0, \ldots$ forever. If the schedule is perturbed, $P = \left(\begin{smallmatrix} 0.1 & 0.9 \\ 0.9 & 0.1 \end{smallmatrix}\right)$, the chain is aperiodic (a state can return in one step), and the same probability oscillates with decreasing amplitude, $1, 0.1, 0.82, 0.24, 0.71, 0.34, \ldots$, towards 0.5.

5.4 Stationary and limiting distributions

Example 5.5 (Long-run market shares)**.** The table of Example 5.2 stops changing at $\pi = (0.5263, 0.3684, 0.1053)$. Starting instead from customers who buy nothing, $\mu_0 = (0, 0, 1)$, the share of O goes $0, 0.30, 0.42, 0.47, \ldots$ and reaches the same 0.5263, as depicted in Figure 5.3. In the long run, our brand accounts for 52.6% of monthly choices, regardless of the current situation. These shares solve $\pi P = \pi$: a population of customers distributed as π is still distributed as π one month later.

Definition 5.4 (Stationary distribution)**.** A probability distribution π on $\mathcal{S}$ is **stationary** (or invariant) for P if $\pi P = \pi$, that is, $\pi_j = \sum_i \pi_i P_{ij}$ for every j. If $X_0 \sim \pi$, then $X_t \sim \pi$ for every t.

Algebraically, π is a left eigenvector of P for the eigenvalue 1, normalized to sum to one; it can be computed by solving the linear system $\pi(P - I) = 0$, $\sum_i \pi_i = 1$.

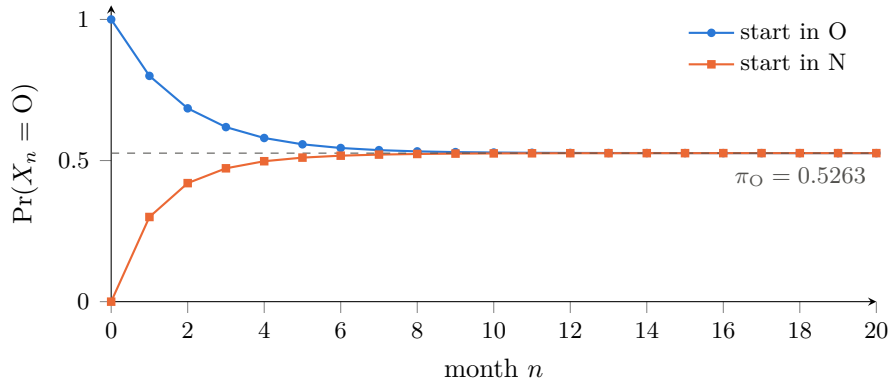


Figure 5.3: Probability that a customer buys our brand after n months, starting from O or from N (Example 5.5). Both curves converge to the stationary share $\pi_O = 0.5263$.

Theorem 5.1 (Convergence of finite Markov chains)**.** *Let P be the transition matrix of an irreducible chain on a finite state space. Then:*

1. *it has a unique stationary distribution π, with $\pi_i > 0$ and $\pi_i = 1/\mathbb{E}_i[\tau_i]$, where τ_i is the time of the first return to i starting from i;*
2. *if, moreover, the chain is aperiodic, then for every initial distribution μ_0, $\mu_0 P^n \to \pi$ as $n \to \infty$; the* ***total variation distance*** $\frac{1}{2}\sum_j |(\mu_0 P^n)_j - \pi_j|$ *decreases geometrically, at a rate governed by the second-largest eigenvalue modulus $|\lambda_2|$ of P.*

For the brand chain, the eigenvalues of P have moduli 1, 0.582 and 0.318. Starting from O, the total variation distance to π is 0.274 after one month, 0.031 after five, 0.002 after ten and 9×10^{-6} after twenty: each month removes about 42% of the remaining distance $(1 - 0.582)$. The mean time between purchases by a customer of our brand is $1/\pi_O = 1.90$ months (a simulation over 200 000 months yields 1.899). In other words, aperiodicity matters. The alternating chain of Example 5.4 has the stationary distribution $(0.5, 0.5)$ but never converges to it, while its perturbed version does. On infinite or continuous state spaces, existence of π also requires **positive recurrence** (finite expected return times). The conclusions are otherwise the same.

Example 5.6 (A chain on a continuous state space)**.** Weekly sales deviations from their seasonal level often follow an **autoregressive** chain $X_{t+1} = \rho X_t + \varepsilon_t$ with $\varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$ and $|\rho| < 1$: the adstock recursion of Section 1.3.1 with random shocks. The transition kernel is $K(x, \cdot) = \mathcal{N}(\rho x, 1)$. If $X_t \sim \mathcal{N}(0, v)$, then $X_{t+1} \sim \mathcal{N}(0, \rho^2 v + 1)$, which is the same distribution when $v = 1/(1-\rho^2)$: the stationary distribution is $\mathcal{N}(0, 1/(1-\rho^2))$. From $X_0 = 20$, the expected value $20\rho^n$ is 0.02 after 10 steps for $\rho = 0.5$, but still 6.97 after 10 steps and 0.10 after 50 for $\rho = 0.9$: the closer ρ is to one, the longer the chain remembers its starting point. Simulations of 200 000 steps give stationary variances of 1.335 ($\rho = 0.5$, exact 1.333) and 5.397 ($\rho = 0.9$, exact 5.263).

5.5 Ergodic theorems

Example 5.7 (One customer's long-run revenue)**.** A customer spends 60 euros per month on our brand. Following a single simulated customer, the share of months with O is 0.42 over the first year, 0.38 over ten years, 0.54 over a century, and 0.530 over 1 000 years, compared with

the stationary share 0.526. The average monthly revenue tends to $60\,\pi_{\mathrm{O}} = 31.58$ euros. One long trajectory slowly recovers the stationary expectation.

Theorem 5.2 (Ergodic theorem and Markov chain CLT)**.** *Let (X_t) be an irreducible, positive recurrent Markov chain with stationary distribution π, and let h satisfy $\mathbb{E}_\pi|h(X)| < \infty$. Then, for every starting state,*

$$\bar{h}_N = \frac{1}{N}\sum_{t=1}^{N} h(X_t) \longrightarrow \mathbb{E}_\pi[h(X)] \quad \textit{with probability one.} \tag{5.3}$$

If, in addition, the chain converges fast enough (geometric ergodicity) and $\mathbb{E}_\pi[h(X)^2] < \infty$, then $\sqrt{N}(\bar{h}_N - \mathbb{E}_\pi h) \to \mathcal{N}(0, \sigma^2\tau)$, where $\sigma^2 = \mathrm{Var}_\pi(h(X))$ and

$$\tau = 1 + 2\sum_{k=1}^{\infty} \rho_k \tag{5.4}$$

*is the **integrated autocorrelation time**, with ρ_k the correlation between $h(X_t)$ and $h(X_{t+k})$ under stationarity.*

Theorem 5.2 is the Markov-chain version of the LLN and the CLT of Proposition 1.2: one trajectory replaces independent draws, but each draw is worth only $1/\tau$ of an independent one. The **ESS** of N correlated draws is N/τ, and the Monte Carlo standard error is $\sigma\sqrt{\tau/N} = \sigma/\sqrt{\mathrm{ESS}}$. The naive standard error $\sigma/\sqrt{N}$, which ignores the correlation, is too small.

Example 5.8 (Correlation costs draws)**.** For the stationary AR(1) chain of Example 5.6, $\rho_k = \rho^k$ (Figure 5.4), so $\tau = (1+\rho)/(1-\rho)$ and the ESS of N steps is $N(1-\rho)/(1+\rho)$. Averaging $N = 10\,000$ steps, repeated 500 times:

ρ	**Spread of the 500 averages**	**Naive SE**	**Correct SE**	ESS
0	0.0100	0.0100	0.0100	10 000
0.5	0.0206	0.0116	0.0200	3 333
0.9	0.0970	0.0229	0.1000	526
0.99	1.0191	0.0701	1.0000	50

With $\rho = 0.99$, ten thousand correlated steps are worth fifty independent draws, and the naive standard error understates the real one fourteen-fold.

5.6 Reversibility and detailed balance

Example 5.9 (Active sessions on the website)**.** The number of visitors currently on a small website (0 to 5) goes up by one each second with probability 0.3 (an arrival, if fewer than 5), down by one with probability 0.4 (a departure, if at least one), and stays the same otherwise. In such a **birth–death chain**, the flow of probability from i to $i+1$ is balanced by the flow back: the stationary distribution $\pi = (0.304, 0.228, 0.171, 0.128, 0.096, 0.072)$ satisfies $\pi_i \times 0.3 = \pi_{i+1} \times 0.4$ for every i. The brand chain is not balanced pair by pair: $\pi_{\mathrm{O}} P_{\mathrm{OK}} = 0.0789$ customers per month switch from O to K, but only $\pi_{\mathrm{K}} P_{\mathrm{KO}} = 0.0737$ switch back; the difference circulates through N.

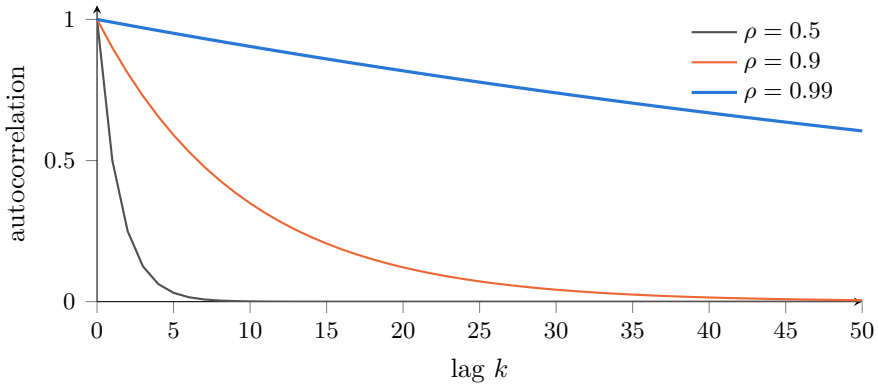


Figure 5.4: Autocorrelation $\rho_k = \rho^k$ of a stationary AR(1) chain at lag k. The slower the decay, the larger the integrated autocorrelation time τ and the smaller the effective sample size.

Definition 5.5 (Detailed balance and reversibility)**.** A transition matrix P satisfies **detailed balance** with respect to a distribution π if

$$\pi_i P_{ij} = \pi_j P_{ji} \qquad \text{for all } i, j \in \mathcal{S}. \tag{5.5}$$

A chain started from π that satisfies detailed balance is **reversible**: the reversed sequence $(X_N, X_{N-1}, \dots, X_0)$ has the same distribution as $(X_0, \dots, X_N)$.

Proposition 5.2 (Detailed balance implies stationarity)**.** *If P satisfies detailed balance with respect to π, then π is stationary for P.*

Proof. For every j, $\sum_i \pi_i P_{ij} = \sum_i \pi_j P_{ji} = \pi_j \sum_i P_{ji} = \pi_j$, because the rows of P sum to one. □

Detailed balance is sufficient but not necessary for stationarity, as the brand chain shows. It is, however, a *local* condition, which involves only pairs of states. This makes it the natural tool for the reverse problem of MCMC: given a target π, *design* a transition matrix that leaves it stationary.

Example 5.10 (A chain whose stationary distribution is the Chapter 1 posterior)**.** Place the conversion rate on a grid of 40 cell midpoints $\theta_j = 0.005j - 0.0025$ covering $(0, 0.2)$, with target weights $\pi_j \propto \theta_j^{13}(1-\theta_j)^{225}$, the Beta$(14, 226)$ density of Section 1.2.1. From state j, propose one of the neighbors $j \pm 1$ with probability $1/2$ each and *accept* the move with probability $\min\{1, \pi_{j'}/\pi_j\}$; otherwise stay at j. For neighbors j and j' with $\pi_{j'} \le \pi_j$,

$$\pi_j P_{jj'} = \pi_j \cdot \tfrac{1}{2} \cdot \frac{\pi_{j'}}{\pi_j} = \tfrac{1}{2}\pi_{j'} = \pi_{j'} \cdot \tfrac{1}{2} \cdot 1 = \pi_{j'} P_{j'j},$$

so detailed balance holds and π is stationary by Proposition 5.2. The transition matrix only uses *ratios* $\pi_{j'}/\pi_j$: the normalizing constant $B(14, 226)$ is never needed. Simulated for 50 000 steps from $\theta = 0.0025$, far in the tail, and after discarding the first 1 000 steps, the chain gives a posterior mean of 0.0582 (exact 0.0583) and $\Pr(\theta > 0.05 \mid \mathbf{y}) \approx 0.696$, against 0.690 for the grid target and 0.6897 for the exact posterior (Figure 5.5). This is the **Metropolis algorithm** on a finite state space; Chapter 6 extends it to continuous parameters.

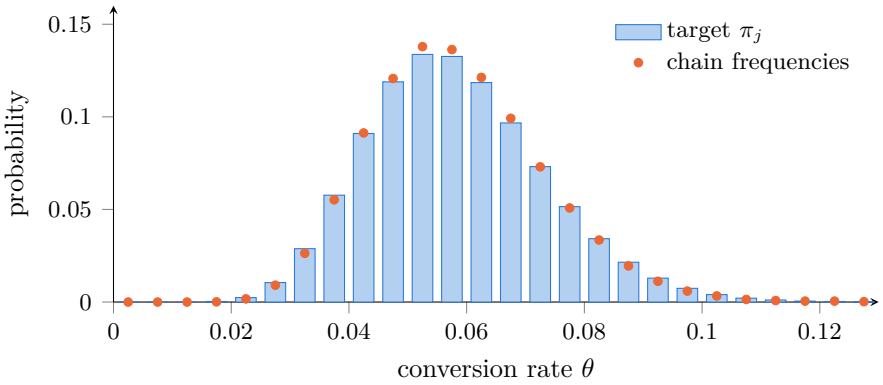


Figure 5.5: Metropolis chain on a grid of conversion rates (Example 5.10): target weights proportional to the Beta(14, 226) density (bars) and frequencies of the visited states (dots).

Summary

Takeaways

- A Markov chain moves from state to state with probabilities that depend only on the current state; its n-step behavior is given by matrix powers, $\mu_n = \mu_0 P^n$ (Chapman–Kolmogorov).
- States are transient or recurrent, and classes may be periodic; absorbing chains are analyzed with the fundamental matrix $N = (I - Q)^{-1}$.
- An irreducible, aperiodic finite chain forgets its starting point: $\mu_n \to \pi$, the unique solution of $\pi P = \pi$, at a geometric rate set by $|\lambda_2|$.
- Time averages along one trajectory converge to stationary expectations (ergodic theorem), but correlated draws are worth fewer independent ones: ESS $= N/\tau$ with $\tau = 1 + 2\sum_k \rho_k$.
- Detailed balance, $\pi_i P_{ij} = \pi_j P_{ji}$, implies stationarity, involves only ratios of π, and is the design principle of MCMC.

Further reading

Norris (1998) develops the theory of discrete Markov chains, including recurrence, convergence, and reversibility, in its first chapter; Ross (2014) gives an applied treatment with many examples, including absorbing chains and the fundamental matrix. Markov chains on general state spaces, with the ergodic theorem and the Markov chain CLT used in simulation, are covered in Chapter 6 of Robert and Casella (2004) and Chapter 4 of Gamerman and Lopes (2006).